

ADAM PARKOSIDIS

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Anton Pannekoek Institute for Astronomy
University of Amsterdam
Amsterdam, North Holland, The Netherlands

EDUCATION

Master of Science <i>Physics and Astronomy</i> ' <i>Cum Laude</i> ' University of Amsterdam-Vrije Universiteit Amsterdam, The Netherlands	2021 – 2023
Thesis: <i>Mass Transfer in Hierarchical Triple Systems: The Case of Roche-Lobe filling Outer Star</i>	
Bachelor of Science <i>Physics</i> Aristotle University of Thessaloniki, Greece	2013 – 2020
Thesis: <i>Dynamic Perturbations in Protoplanetary Cosmic Dust Discs in the Presence of a Giant Planet: Possible Detection with ALMA</i>	

PUBLICATIONS

Reframing the wide eccentric binary problem: Eccentricity as a probe of mass-transfer physics A. Parkosidis, S. Toonen, E. Laplace, V. Schaffenroth	2026, accepted ApJ
Rethinking mass transfer: a unified semianalytical framework for circular and eccentric binaries. II. Orbital evolution due to non-conservative mass transfer A. Parkosidis, S. Toonen, E. Laplace, F. Dosopoulou	2026, A&A
Rethinking mass transfer: a unified semianalytical framework for circular and eccentric binaries. I. Orbital evolution due to conservative mass transfer A. Parkosidis, S. Toonen, F. Dosopoulou, E. Laplace	2026, A&A
Episodic outbursts during brown dwarf formation A. Parkosidis, D. Stamatellos, B. Riaz	2025, A&A

SCIENTIFIC EXPERIENCE

PhD Candidate in Astrophysics Anton Pannekoek Institute, UvA, The Netherlands	2023 – Present
Technical Support Scientist ASTRON, Dwingeloo, The Netherlands	2023
- Supported LOFAR Unified Calculator for Imaging (LUCI) tool development by testing, debugging, and integrating new Python functionalities - Developed observation and processing templates utilizing LOFAR's Telescope Manager Specification System following proposal approval - Authored documentation and user guides on instrument capabilities and operational procedures - Performed quality control of data products generated by the Long-Term Archive (LTA) pipelines, producing reports and performance summaries	
Research Intern in Computational Astrophysics Jeremiah Horrocks Institute, UCLan, United Kingdom	2021
- Applied the PHANTOM smoothed particle hydrodynamics (SPH) code for large-scale numerical simulations of protoplanetary disks - Utilized the UCLan High-Performance Computing (HPC) cluster for parallel execution and resource-optimized SPH model runs - Processed, analyzed, and visualized simulation data using Python (NumPy, pandas, matplotlib, seaborn) and SPLASH	
Space Debris Data Analyst SpaceDot, Thessaloniki, Greece	2020 – 2023
Collaborated with 80+ engineers and scientists to integrate debris mitigation data into mission design	

- Performed orbit propagation and collision analyses with ESA's DRAMA software to ensure compliance with international space debris mitigation standards
- Developed Python scripts for automating orbital simulations, data analysis, and visualization of on-orbit collision risk scenarios
- Contributed technical content to the System Design and Mission Requirements (SDMR) and Orbital Collision Risk Assessment (OCRA) documents for the Critical Design Review (CDR) phase

FUNDING & GRANTS

Leids Kerkhoven-Bosscha Fonds (LKBF)

Years granted: 2024, 2025, 2026

TALKS

'When Mass Transfer Makes Orbits Eccentric: Connecting Theory and Observation'	2026
European Astronomical Society	Lausanne, CHE
'Eccentricity as a Probe of Mass-transfer Physics'	2026
Invited talk: Institute of Astronomy	KU Leuven, BEL
'Mass Transfer in Eccentric Binaries: A Unified Framework & New Insights'	2025
Binary Stars in a New Era	Lijiang, CHN
'Mass Transfer in Eccentric Binaries: A Unified Framework & New Insights'	2025
Netherlands Astronomy Conference	Nijmegen, NL

TECHNICAL SKILLS

Operation Systems: Linux (Ubuntu), Microsoft Windows
Programming: Python, Bash, C, F77/F95
Machine Learning & Deep Learning: PySR, Keras, Pandas, PyTorch, Scikit-Learn, TensorFlow
Data Visualization: Matplotlib, Seaborn, Plotly, gnuplot
Document Creation: Microsoft Office Suite, LibreOffice Suite, Google Suite, LaTeX
Other Technologies: Smoothed Particle Hydrodynamics (SPH), Git

SCIENTIFIC COMPUTING

SPH codes: PHANTOM, SEREN
Stellar Evolution codes: MESA
N-body codes: SWIFT, HUAYNO
Python Packages: NumPy, SciPy, SymPy, DISKLAB, Astropy, PICASO
Space Debris Risk Assessment & Mitigation Analysis: DRAMA
Other software: AMUSE, Mathematica

TEACHING EXPERIENCE & STUDENT SUPERVISION

University Lectures	2026
• 'Convection' (M.Sc. course: 'Structure & Evolution of Stars')	
Summer projects	2024
• Magdalena Andrea Vilaxa Campos "Modeling Mass Transfer Between Stars" (MSc level)	
Teaching Assistant	2022 – 2025
Anton Pannekoek Institute, UvA, The Netherlands	
• 'Atmospheres & Radiative Transfer' (M.Sc. course, 4 times)	
• 'Structure & Evolution of Stars' (M.Sc. course, 2 times)	
• 'Disks & Accretion' (M.Sc. course)	
AUTh, Greece	2018
• 'Observational Astronomy' (BSc course)	
Private Tutor	2018 – 2019
Thessaloniki, Greece	
• Tutored and prepared secondary education students for their final exams in the school subjects of Physics and Math	

TRAININGS & SCHOOLS

Binary compact objects on eccentric orbits and their gravitational-wave signatures Corfu, Greece	2026
ICESUN Summer School 2025: Binary Stars and Compact Objects Yunnan Observatories, China	2025
MESA Summer School KU Leuven, Belgium	2025
Nordic Winter School on Multimessenger Astrophysics Skeikampen, Norway	2024
Mastering your PhD University of Amsterdam, The Netherlands	2024

COLLABORATION MEMBERSHIPS

The Binarity at LOw Metallicity (BLOeM)	2026 – Present
Laser Interferometer Space Antenna (LISA)	2025 – Present

LANGUAGES

Greek: Native
English: Fluent
Dutch: Elementary
German: Elementary

VOLUNTEERISM

Guide and telescope operator Observatory of AUTH	2018 – 2019
Operated the optical telescope at the Aristotle University of Thessaloniki Observatory during visits by primary, junior high, and high school students.	